Developing ECG Signal Processing Techniques for Noise Suppression in Wearable Devices

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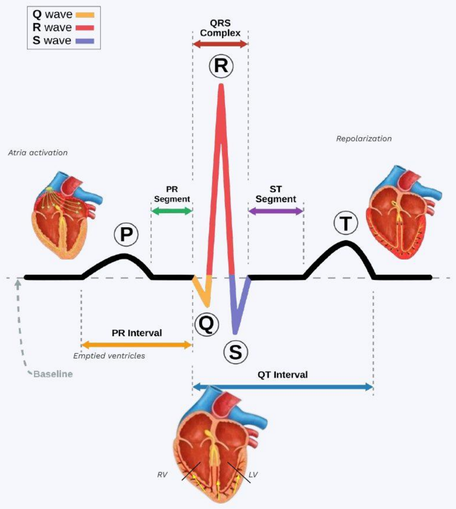
Introduction

- Heart disease is leading cause of death, with 805,000 heart attacks annually in the U.S.
- Caused by oxygen-rich blood blockage due to artery clots.
- Current Methods:
  1) Blood tests are the only method that are 100% accurate
  2) Biomarkers can only be tested as early as a few hours after the heart attack
  3) Continuous ECG (Electrocardiogram) systems
     a) Require strict placement + multiple nodes
     b) Only available in hospitals

How are heart attacks detected?

- Heart beats generate an electrical impulse that can be detected with electrodes that are attached to our skin.
- Electrocardiogram (ECG) detect electrical impulses

- Read through QRS complex
  - Q wave → depolarization of the interventricular septum (breathing)
  - R wave → depolarization of main mass
  - S wave → final depolarization



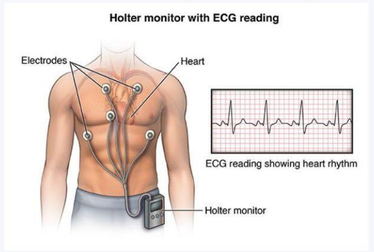
Goal: Develop a cost-effective, wearable ECG system for continuous heart monitoring and heart attack detection and prevention

Research question: Which biomarkers on ECG help detect a heart attack BEFORE it occurs?

Limitations/issues in current technology

Hospital ECGs

Expensive, one-time test (NIH 2024)



Blood tests

Used AFTER heart attack happened (NIH 2024)



Smart watches

Inaccurate + noise



Figure 1: Circuit schematic

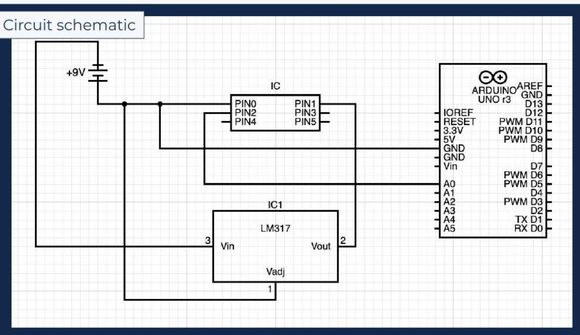


Figure 2: Circuit design

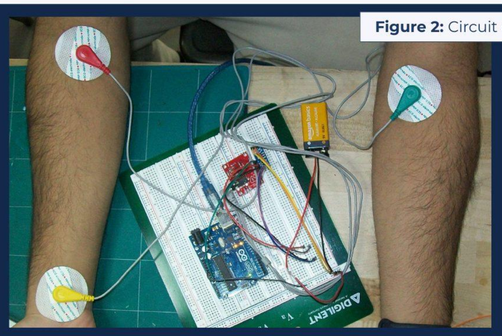




Figure 3: Digital Waveform

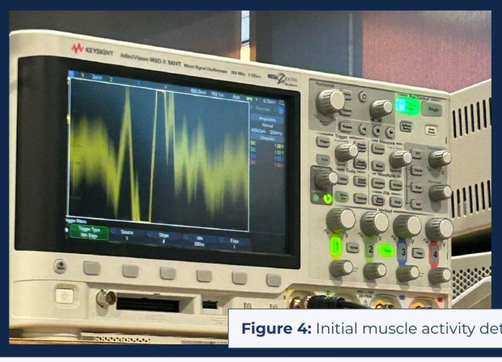


Figure 4: Initial muscle activity detection

Components Used

- AD8232 (Biopotential measurement chip)
- Arduino UNO
- Electrode patches with 3 connector cables
- 9V Battery
- 3.3V Regulator Module





Methodology

1) Collect & Display signal:
2) Solder chip pins to AD8232
3) Attach jumper wires to all the pins
4) Connect GND, 3.3v, Output to A0, LO- to 11, and LO+ to 10 on Arduino
5) Attach the connector cable jack to the ECG Biomedical Connector on AD
6) Code A0 pin to read signal input
7) Analyze waveforms displayed wrt. to time

Future Steps

- **Design for miniaturization and wearability**
  - Optimize electrode placement for long-term attachment (use high quality gel and/or adhesive)
  - Focus on comfort by using flexible materials
- **Signal processing**
  - Converting signal to dataframe object
  - Work on noise reduction from motion and EMG interference
  - Further analysis on dataframe utilizing ML algorithms
- **Optimize detection for various cardiovascular diseases**
  - Train models to recognize certain patterns
  - Research ST elevations, T wave inversions, arrhythmias, etc.

References

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2. Aatik. (n.d.). DIY Homemade ECG Machine using AD8232. YouTube. https://www.youtube.com/watch?v=1EGhAKdqAag
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